CS-471: Machine Learning

Week 3

Instructor: Dr. Hashir Kiani

Classroom Etiquette and Participation Guidelines

- Treat everyone with respect and courtesy at all times
- Speak only when it is your turn and raise your hand before contributing to the discussion
- Do not interrupt others while they are speaking
- Join the class on time
- Keep mobile phones on silent during the lecture

Lecture Outline

- Recap of Previous Lecture
- Logistic Regression
- Quiz

Recap of Previous Lecture

- How do we split a dataset for evaluation?
- What is training set used for?
- What is validation set used for? Testing set?
- What is MSE and RMSE?
- What is Mean Absolute Error?
- What is Mean Absolute Percentage Error?
- What is R-Squared? How do we find it?
- What is binary classification? Examples?
- What is the hypothesis class in binary linear classification?

Recap of Previous Lecture

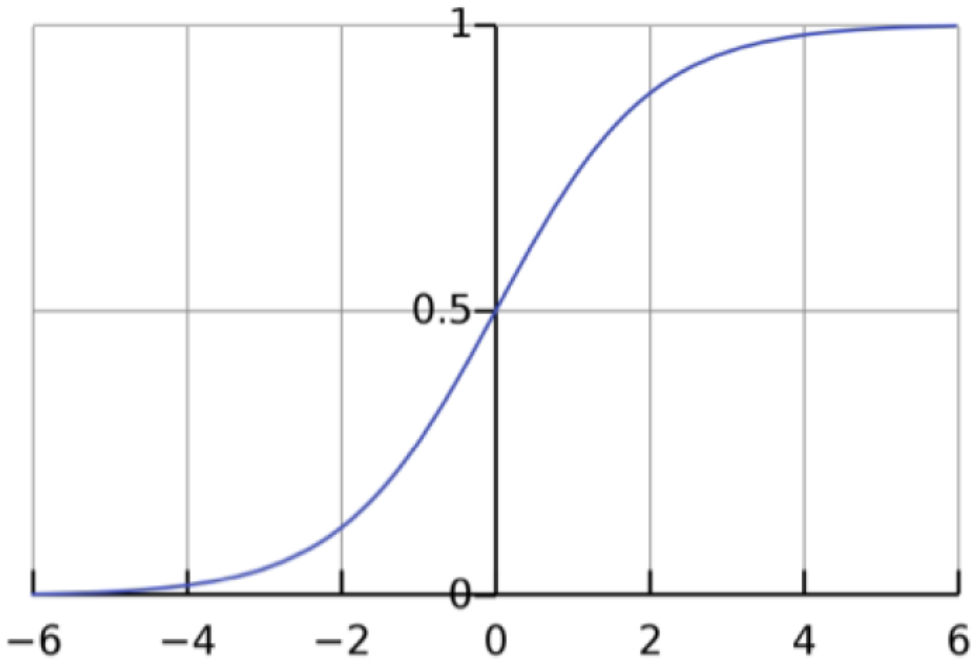
- How do we define the decision boundary in binary linear classification?
- What is score? Margin?
- What is Zero-One Loss?
- What is the problem with zero-one loss?
- What is hinge loss?
- What is the gradient of hinge loss?
- Give the equation of gradient descent for binary linear classification using hinge loss.

Logistic Regression – Another Linear Classifier

- Hypothesis class: $h(\boldsymbol{x}) = g(\boldsymbol{w}^T \phi(\boldsymbol{x}))$

Where $\mathrm{g}(z) = \dfrac{1}{1+e^{-z}}$ (Sigmoid/Logistic Function)

- Output interpreted as probability of positive label.
- $g(z) > 0.5$ means positive label
- $g(z) \leq 0.5$ means negative label



Logistic Regression – Loss Function

- Logistic loss also known as Binary Cross Entropy Loss:

$$Loss\big(y, h(\boldsymbol{x})\big) = -ylog\big(h(\boldsymbol{x})\big) - (1-y)\log(1-h(\boldsymbol{x}))$$

- Training loss in this case:

$$TrainLoss(\boldsymbol{w}) = -\frac{1}{N}\sum_{i=1}^{N}[y_i log\big(h(\boldsymbol{x_i})\big) + (1-y_i)\log(1-h(\boldsymbol{x_i}))]$$

Logistic Regression – Optimization Algorithm

- Gradient Descent:

$$\boldsymbol{w}_{new} = \boldsymbol{w}_{old} - \alpha \nabla_{\boldsymbol{w}} TrainLoss(\boldsymbol{w})$$

$$\nabla_{\boldsymbol{w}} TrainLoss(\boldsymbol{w}) = \frac{1}{N} \sum_{i=1}^{N} [(h(\boldsymbol{x_i}) - y_i)\boldsymbol{x_i}]$$

$$\boldsymbol{w}_{new} = \boldsymbol{w}_{old} - \frac{\alpha}{N} \sum_{i=1}^{N} [(h(\boldsymbol{x_i}) - y_i)\boldsymbol{x_i}]$$

Linear Classification Example

- Training data:

x	y
$[0 \quad 2]$	1
$[-2 \quad 0]$	1
$[1 \quad -1]$	-1

Questions?